Neighbours and Orientation: Out-of-Distribution Detection with Frozen Diffusion Features

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Abstract

One pass of a frozen diffusion denoiser yields label-free features for out-of-distribution (OOD) detection, and a k-nearest-neighbour (kNN) rule on them already beats likelihoods and score statistics of the same model. We show that these features hold a second notion of in-distribution (ID) conformity that kNN does not use. kNN measures closeness to training images along the high-variance directions of the features; a small head trained to predict the rotation of ID images reads out whether an image carries the *canonical frame* of the ID data, which way is up. On ImageNet-64 and CIFAR-100 a half turn moves an image by less than its distance to the neighbours the kNN rule uses (0.30 of it on ImageNet-64), so a distance barely sees the frame there, whereas a small head reads it out of the same statistics. Both notions detect distribution shift, but on different images. Bayes' rule for the rotation mixture, $\log p = \log \bar{p} + \log P(\mathrm{up} \mid \mathbf{x}) + \log 4$ with $\bar{p}$ the rotation-averaged density, suggests how to combine them: Upright kNN adds the log-posterior of the upright pose to the kNN log-density, with the unit weight of the identity rather than a tuned one. Under a protocol frozen before the final evaluation, on six ID datasets it reaches 93.4% mean AUROC on natural OOD sets (its kNN term alone 92.5%, a kNN rule tuned per dataset 92.6%, the best of 14 prior detectors 85.5%; CSI, trained on the ID pixels, is within two points on four of the five 32×32 datasets and collapses on CIFAR-100, 92.2% against our 96.1% over the five) at one network evaluation per image, and keeps its margin, without refitting, on the full OpenOOD v1.5 ImageNet-1K benchmark at 64×64.

1 Introduction

Out-of-distribution (OOD) detection asks whether a test input shows something that the training data do not. Without labels or OOD data, a generative model of the in-distribution (ID) data is the natural tool, but the likelihoods of deep generative models track image complexity more than novelty [Nalisnick *et al.*, 2019a; Serrà *et al.*, 2020], and detectors built on the score fields, sampling paths, reconstructions or posterior statistics of diffusion models [Mahmood *et al.*, 2021; Graham *et al.*, 2023; Liu *et al.*, 2023; Heng *et al.*, 2024; Kamkari *et al.*, 2024; Rouzoumka *et al.*, 2026b; Shoushtari *et al.*, 2026] need 10 to more than 10,000 network evaluations per image and swing between ID datasets (the likelihood ranges from 33% to 99% AUROC in Table 2). A single denoiser pass is cheaper and stronger: CFS [Rouzoumka *et al.*, 2026a] scores pooled channel statistics of its activations against ID images and beats these detectors, and a kNN rule [Sun *et al.*, 2022] on such features, tuned per ID dataset, reaches 92.6% in our evaluation.

Is closeness to training images all that these features say about ID conformity? Self-supervised detectors have long used a second signal, the prediction of the rotation applied to an image [Golan and El-Yaniv, 2018; Hendrycks *et al.*, 2019], and CSI [Tack *et al.*, 2020] scores a test image by its similarity to the nearest training instance plus the confidence of a rotation head. Because these methods train the representation on ID pixels, they do not ask whether the two are different signals or the same signal read twice. We ask the question inside one frozen representation, the features of a single denoiser pass.

They are different, and the difference is geometric (Section 5). A kNN rule measures *neighbourhood conformity*: closeness to some training image along the directions in which ID features vary most. A head trained to predict the rotation of ID images measures *frame conformity*: whether the image carries the canonical frame of the ID data, the orientation convention that says which way is up. On ImageNet-64 and CIFAR-100 a half turn moves an image by less than its distance to its neighbours, so the kNN rule barely notices upside-down images there; the head does. Both notions detect distribution shift, but on OOD images their scores are weakly correlated and flag different images: unusual appearance for the first, unusual viewpoint or composition for the second.

The two notions are the two factors of Bayes' rule for the rotation mixture. With $\bar{p}$ the density averaged over the four rotations by multiples of 90°, $\log p = \log \bar{p} + \log P(\mathrm{up} \mid \mathbf{x}) + \log 4$ holds for every image (Proposition 1): a density that does not see orientation can at best estimate $\log \bar{p}$, and the missing factor is the posterior that the image is upright. Upright kNN therefore adds the log-posterior of a rotation head to the kNN log-density (Figure 1); the relative weight of

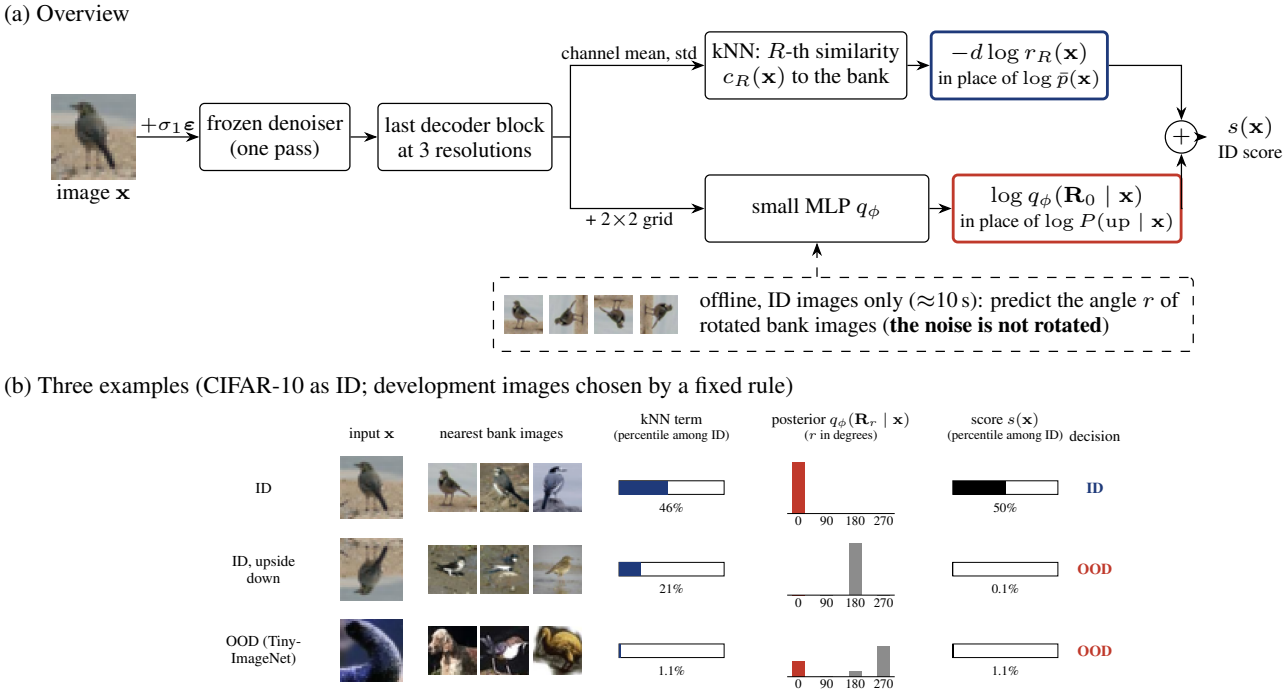


Figure 1: Two notions of ID conformity read from one denoiser pass. (a) The kNN term measures closeness to the nearest bank images (neighbourhood conformity); the orientation term, whether a small rotation head recognises the canonical frame of the ID data in the image (frame conformity); Upright kNN adds the two, Eq. (2). (b) From top: the ID image with the median score, the same image upside down, and the median-score Tiny-ImageNet image. Percentile: fraction of ID images with a lower value; decision: threshold keeping 95% of ID images. The flipped bird keeps a bird as its nearest bank image and its kNN term stays at the 21st percentile, but its orientation posterior moves to 180°.

the two terms is not tuned but taken from the identity, because the kNN log-density is $-d \log r_R$ with d the intrinsic dimension of the features. We do not claim that the score estimates $\log p$. Upside-down ID images serve only as a probe of frame sensitivity, since for left–right symmetric data a vertical flip changes the frame factor alone (Proposition 2); the target is distribution shift on natural OOD datasets.

The kNN rule on these features is itself a strong detector; the orientation term adds modestly on average and most where the frame is informative. Under a protocol frozen before the held-out images were evaluated, once, Upright kNN reaches 93.4% mean AUROC on 47 natural ID/OOD pairs, 0.9 points above its own kNN term (92.5%; 28 wins, 1 loss) and 7.9 points above the best of 14 prior detectors (85.5%; thirteen of them use the same diffusion models), at one network evaluation per image. Our contributions are:

- A characterisation of two notions of ID conformity in frozen diffusion features: where each lives in feature space, what each responds to, how they relate per image, and when the frame notion fails (Section 5).
- Upright kNN, a single-pass, label-free detector that combines them with the unit weight suggested by Bayes' rule, which lies on the plateau of a weight sweep, and an account of why the kNN density, not the likelihood, is the partner of the frame term (Section 3).
- A frozen-protocol evaluation against 14 prior detectors and CSI on six ID datasets and on OpenOOD ImageNet-1K at 64, 256 and 512 pixels, with failure cases reported (Section 6).

2 Related Work

Likelihoods and statistics of generative models. Responses to the likelihood paradox [Nalisnick *et al.*, 2019a] include likelihood ratios [Ren *et al.*, 2019], typicality tests [Nalisnick *et al.*, 2019b], complexity corrections [Serrà *et al.*, 2020], ensembles of statistics [Morningstar *et al.*, 2021] and local intrinsic dimension [Kamkari *et al.*, 2024]. Diffusion-specific detectors use multiscale score norms (MSMA) [Mahmood *et al.*, 2021], reconstruction or inpainting errors (DDPM-OOD, LMD) [Graham *et al.*, 2023; Liu *et al.*, 2023], sampling-path statistics (DiffPath) [Heng *et al.*, 2024], the consistency of the posterior mean under transformations (GEPC) [Rouzoumka *et al.*, 2026b] and posterior-covariance eigenvalues (EigenScore) [Shoushtari *et al.*, 2026]; we run all of them.

Feature-based detection. With supervised classifiers, Mahalanobis [Lee *et al.*, 2018] and deep kNN [Sun *et al.*, 2022] scores on penultimate features are strong baselines [Yang *et al.*, 2024; Zhang *et al.*, 2023]. Diffusion denoisers provide features without labels [Baranchuk *et al.*, 2022; Tang *et al.*, 2023]; CFS [Rouzoumka *et al.*, 2026a] pools channel statistics of one encoder and one decoder block and scores them with a diagonal Gaussian or a kNN head; our kNN term uses the same kind of features. Encoders pretrained on external data, such as CLIP [Radford *et al.*, 2021] and

DINOv2 [Oquab *et al.*, 2024], give strong zero-shot detectors [Ming *et al.*, 2022] in a different regime, the ID data defined relative to a larger corpus; we compare detectors that use the ID images and a model trained on them.

Transformation prediction. Predicting the rotation or another geometric transformation of an ID image [Gidaris *et al.*, 2018] gives an anomaly score [Golan and El-Yaniv, 2018; Hendrycks *et al.*, 2019; Bergman and Hoshen, 2020]. CSI [Tack *et al.*, 2020] trains a contrastive representation with rotated images as negatives and scores a test image by its similarity to the nearest training instance plus the confidence of a rotation head; SSD [Sehwag *et al.*, 2021] uses Mahalanobis distances in a contrastive representation. These methods train a network on ID pixels for hundreds of epochs; we train no representation and show that a frozen generative one already separates the two kinds of evidence.

3 Method

Setting. We are given ID training images and a diffusion model trained on them, and use its denoiser $D(\mathbf{y};\sigma)$ with no labels and no OOD data; a detector assigns a score $s(\mathbf{x})$ that is higher for ID-like images.

Features from one pass. We add noise $\sigma_1\boldsymbol{\varepsilon}$ to $\mathbf{x}$, with $\sigma_1 = 0.01$ for all models and one noise image $\boldsymbol{\varepsilon}$ shared by all images, and pass $\mathbf{x} + \sigma_1\boldsymbol{\varepsilon}$ through the denoiser once. From the output of the last decoder block at full, half and quarter resolution we take the channel-wise spatial mean and standard deviation, $\mathbf{g}(\mathbf{x})$ ($2C$ values per block with C channels), and the same statistics over each cell of a 2×2 grid, $\mathbf{q}(\mathbf{x})$ ($8C$ values per block). The kNN term uses $\mathbf{g}$ only; the head uses $\mathbf{g}$ and $\mathbf{q}$. The same features are computed for a bank of ID training images $\mathbf{b}_1, \ldots, \mathbf{b}_N$.

3.1 Neighbourhood Conformity: a kNN Density

We normalise each block's part of $\mathbf{g}$ to unit length. Let $c_R(\mathbf{x})$ be the R-th largest similarity between $\mathbf{x}$ and the bank images (cosine similarity averaged over the three blocks) and $r_R(\mathbf{x}) = \sqrt{2\{1 - c_R(\mathbf{x})\}}$ the corresponding chordal distance. The kNN density estimate [Loftsgaarden and Quesenberry, 1965] in a space of dimension d is proportional to $r_R(\mathbf{x})^{-d}$, so

$$\log \hat{p}_{\mathrm{nn}}(\mathbf{x}) = -d \log r_R(\mathbf{x}) + \mathrm{const}, \tag{1}$$

with d the maximum-likelihood intrinsic dimension [Levina and Bickel, 2004] of the bank features in the same metric (20 neighbours of 3,000 bank points; d is 14 to 25 across our ID datasets, and 5 or 50 neighbours change it by -11 to $+39$%, Section 3.3). R is chosen per ID dataset on selection data among ten values from 1 to 1000 (Section 4; $R = 2$, 5, 5, 5, 1000 and 1 for the six datasets in the order of Table 2, with a bank of 50,000 images, 10,000 for ImageNet-64). On its own, Eq. (1) ranks images as the kNN rule with similarity c_R does.

3.2 Frame Conformity: an Upright Posterior

Let $\mathcal{G} = \{\mathbf{R}_0, \mathbf{R}_{90}, \mathbf{R}_{180}, \mathbf{R}_{270}\}$ be the rotations of the image plane by multiples of 90°. A small MLP $q_\phi(\mathbf{R}_r \mid \mathbf{x})$, one hidden layer of 512 ReLU units and four outputs, reads $[\mathbf{g}, \mathbf{q}]$ of the three blocks, standardised per feature on the bank. We hold out 5,000 bank images (the kNN bank keeps all of them), rotate the others by each $\mathbf{R}_r \in \mathcal{G}$ and minimise the cross-entropy of predicting r (about 10 s; recipe in the supplement); because the four angles are equally likely, the minimiser is the posterior of the rotation given the features. The *noise is not rotated*: the inputs are $\mathbf{R}_r\mathbf{b}_i + \sigma_1\boldsymbol{\varepsilon}$, since a rotated shared $\boldsymbol{\varepsilon}$ would let the head read the angle from the noise (on ImageNet-64 held-out accuracy then rises from 77.3% to 91.6%, but the six-dataset mean flip AUROC of the score on the development images falls from 86.4% to 78.8%). A temperature fitted on the held-out images by negative log-likelihood calibrates the posterior [Guo *et al.*, 2017]; held-out accuracy is 77–100% across the six datasets.

3.3 Combining the Two Notions

Let $\bar{p}(\mathbf{x}) = \frac{1}{4}\sum_{\mathbf{G}\in\mathcal{G}} p(\mathbf{G}\mathbf{x})$ be the density averaged over the four rotations. Rotating ID images by an angle drawn uniformly from $\mathcal{G}$ gives a mixture with density $\bar{p}$, because rotations permute pixels (they preserve volume) and $\mathcal{G}$ is closed under inversion; by Bayes' rule the posterior that $\mathbf{x}$ was not rotated is $P(\mathrm{up} \mid \mathbf{x}) = p(\mathbf{x})/\{4\bar{p}(\mathbf{x})\}$.

Proposition 1. *For every image* $\mathbf{x}$ *with* $p(\mathbf{x}) > 0$, $\log p(\mathbf{x}) = \log \bar{p}(\mathbf{x}) + \log P(\mathrm{up} \mid \mathbf{x}) + \log 4$.

The first term does not depend on orientation, since $\bar{p}(\mathbf{G}\mathbf{x}) = \bar{p}(\mathbf{x})$; the second states how plausible the present orientation is, given the content. A score that is insensitive to rotation can at best estimate the first term. Our kNN density is not exactly rotation-insensitive (Section 5), and making it so, by adding the four rotations of every bank image to the bank, lowers the AUROC on natural OOD sets (92.6% against 93.4% on the development images, driven by CIFAR-10; supplement); we keep it as it is and add the missing factor:

$$s(\mathbf{x}) = -d \log r_R(\mathbf{x}) + \log q_\phi(\mathbf{R}_0 \mid \mathbf{x}). \tag{2}$$

Both terms are measured in nats, as in Proposition 1, so no relative weight is tuned: d is the exponent of the kNN density, and the temperature of the head is fitted on held-out ID images. Still, s is not an estimate of $\log p$, and we do not use it as one: the kNN term is a density in feature space, its rotation invariance is approximate, and the head is fitted on rotated ID images only. The weight is not a knife edge: with $s_w = -d \log r_R + w \log q_\phi$, the mean AUROC on the development images rises from 92.4% at $w = 0$ to a plateau within 0.1 points of its maximum (93.4%) for w between 1 and 2 and falls again at $w = 4$ (93.0%); another estimate d' of the dimension is $w = d/d'$, so 5 or 50 neighbours ($w \approx 0.75$, 1.1) stay within 0.13 points of it (supplement). The parameter-free weight lands on this plateau, as do fusions that fit a scale on extra ID images (Section 6.2). At test time s costs one denoiser pass, inner products with the bank and a 2-ms head.

Why the kNN density and not the likelihood. Proposition 1 also says what adding $\log P(\mathrm{up} \mid \mathbf{x})$ to the model's own log-likelihood would do: nothing useful, for two reasons. The log-likelihood varies by thousands of nats across ID images (standard deviation 1,600 nats on CIFAR-10) and the frame factor by a few (0.7), so the sum ranks images as the likelihood does; and the likelihood is a density of the oriented

image, so it already contains whatever the model knows of the frame, and adding the factor counts it twice (the kNN term counts it twice only on CelebA and FashionMNIST, where it sees the frame and the AUROC is saturated). The kNN term varies by about 3 nats, the scale of the frame factor, so there the factor can change the ranking (Section 5 confirms the first point).

Upside-down images as a probe. A vertical flip $\mathbf{V}$ equals a horizontal flip $\mathbf{H}$ composed with a half turn, $\mathbf{V} = \mathbf{H}\mathbf{R}_{180}$.

Proposition 2. *If* $p(\mathbf{H}\mathbf{x}) = p(\mathbf{x})$ *for all* $\mathbf{x}$*, then* $\bar{p}(\mathbf{V}\mathbf{x}) = \bar{p}(\mathbf{x})$*, so* $\log p(\mathbf{V}\mathbf{x}) - \log p(\mathbf{x}) = \log P(\mathrm{up} \mid \mathbf{V}\mathbf{x}) - \log P(\mathrm{up} \mid \mathbf{x})$.

The proof uses $\mathbf{G}\mathbf{H} = \mathbf{H}\mathbf{G}^{-1}$ for $\mathbf{G} \in \mathcal{G}$ (supplement). Upside-down ID images keep their content and change only the frame factor, so the AUROC between ID images and their flips measures how much a score depends on the frame. The premise holds approximately for photographs and faces (the kNN term separates ID images from their mirror images at most at 52.6% AUROC on four datasets) but not for SVHN (57.2%) and FashionMNIST (77.0%), where digits and shoes have a handedness, so there the probe also measures content (supplement). We use the probe for this purpose only; upside-down images are not an OOD target.

4 Experimental Setup

ID datasets and models. We use CIFAR-10 [Krizhevsky, 2009] (EDM [Karras *et al.*, 2022]), SVHN [Netzer *et al.*, 2011] (the VP-SDE of Kamkari *et al.* [2024]), CelebA [Liu *et al.*, 2015] (iDDPM [Nichol and Dhariwal, 2021]), CIFAR-100 (the EDM-preconditioned DDPM++ of Chen *et al.* [2024]), ImageNet-64 [Chrabaszcz *et al.*, 2017] (EDM2-XS [Karras *et al.*, 2024]) and FashionMNIST [Xiao *et al.*, 2017] (a U-Net we trained). All models are unconditional; all but the last are public and used unchanged. Images are 32×32 except ImageNet-64.

OOD sets. Each ID dataset is paired with natural OOD datasets (9, 9, 9, 9, 7 and 4 sets): for the 32×32 datasets the other three, Textures, Tiny-ImageNet, the LSUN-FIX and ImageNet-FIX sets of Tack *et al.* [2020], ImageNet32 and Places365; for ImageNet-64 SSB-hard, NINCO, iNaturalist, Textures, OpenImage-O, Places365 and CelebA; for FashionMNIST greyscale digit and character sets. These 47 natural pairs are the target; four are near-OOD (CIFAR-10↔CIFAR-100, SSB-hard, NINCO) and get a mean of their own. Each ID dataset is also paired with its upside-down images (the probe) and, except ImageNet-64, with uniform-noise and constant images.

Protocol. Images are split into three disjoint parts: (i) selection data (ID development images and 6–11 OOD datasets per ID dataset that are never used for evaluation), (ii) development evaluation images (1,000 per set; 2,000 for FashionMNIST) and (iii) final evaluation images (500 per set). The configuration of Upright kNN (R and the block set of the kNN term) was chosen on (i); the form of the score was developed on (ii), which draws on the same OOD datasets as (iii), so the frozen evaluation guards against image-level, not dataset-level, over-fitting; the protocol was frozen before (iii) was evaluated once; results are on (iii) unless stated otherwise. The bank is 50,000 ID training images (10,000 for ImageNet-64, 10 per class). Analyses and comparison methods added later leave the method unchanged.

Comparison methods. With the same diffusion models we run the likelihood (NLL, by the probability-flow ODE [Song *et al.*, 2021]), input complexity with PNG and JPEG2000 codes (IC) [Serrà *et al.*, 2020], the typicality test (TT) [Nalisnick *et al.*, 2019b], a Mahalanobis version of MSMA (MSMA*) [Mahmood *et al.*, 2021], DiffPath [Heng *et al.*, 2024], GEPC [Rouzoumka *et al.*, 2026b], EigenScore [Shoushtari *et al.*, 2026], DDPM-OOD [Graham *et al.*, 2023], LMD [Liu *et al.*, 2023], the likelihood + local-intrinsic-dimension rule of Kamkari *et al.* [2024] (LID+NLL; dual-threshold AUROC, an oracle over threshold pairs) and CFS with its diagonal and kNN heads [Rouzoumka *et al.*, 2026a] (official code, our bank). Trained on the ID training images are a ResNet-18 rotation classifier [Hendrycks *et al.*, 2019] and CSI [Tack *et al.*, 2020] (official code, 1,000 epochs on the bank images of the five 32×32 ID datasets; on our final images it agrees with the published AUROCs within three points, supplement). Finally, the kNN rule on our features, basic (half-resolution block, 10th neighbour) or tuned per ID dataset on (i) among 80 configurations (blocks, noise level, statistic, R).

Metrics. We report AUROC (ID as positive) and the FPR at 95% TPR (FPR95). Upright kNN, both kNN rules, CFS, GEPC and EigenScore are averaged over three noise seeds (the seed standard deviation of a per-dataset mean is at most 0.16 points). For pairwise comparisons we bootstrap ID and OOD images (2,000 replicates) and count a win (loss) when the 95% interval of the AUROC difference excludes zero and the difference is at least 0.05 points; because many pairs are near 100%, we also count the wins and losses of at least one point.

5 Two Notions of ID Conformity

We now look inside the features (Table 1; other fits and the development images give the same picture, supplement).

A rotation barely moves an image relative to its neighbours. In the space of the kNN term, a half turn moves a bank image by 0.30 (ImageNet-64) to 2.55 (FashionMNIST) times its distance to its R-th nearest bank image (0.42 to 2.06 at a fixed rank of 10, in the same order; supplement). Where the ratio is below one, the rotated image stays inside its own neighbourhood, and the kNN term detects upside-down ID images at nearly chance level (ImageNet-64 50.5%, CIFAR-100 56.5%). Across the six datasets the flip AUROC of the kNN term follows the share of the feature variance that the rotation explains (0.7% on ImageNet-64, 18.3% on CelebA, where the kNN term does detect flips; Spearman $\rho = 0.83$ over the six datasets), a quantity that does not depend on R.

On natural images the frame is read out, not measured by a distance. The rotation does not hide in quiet directions of the head's input space: the three directions spanned by the differences between the mean features of the four rotations carry 4.9–20.8% of the total variance, 135–436 times three random directions. But on the four natural-image datasets the shift along them is small compared with the spread of ID im-

ID dataset	Geometry			Upside-down ID images (probe)					Natural OOD sets (mean)			
	$\|\Delta_{180}\|/r_R$	rot. var. %	var. frame	kNN term	PC-3	frame-3	head on $\mathbf{g}$	orient. term	kNN term	orient. term	sum	ρ
CIFAR-10	1.11	4.1	0.17	77.8	61.3	94.6	99.0	99.2	93.7	93.8	95.4	0.18
SVHN	1.06	5.1	0.14	56.1	53.7	60.5	81.7	78.6	99.4	95.7	99.7	0.36
CelebA	1.71	18.3	1.7	99.5	99.7	100.0	100.0	100.0	100.0	100.0	100.0	0.12
CIFAR-100	0.81	1.1	0.08	56.5	52.6	75.2	93.8	95.0	85.0	72.9	85.5	0.10
ImageNet-64	0.30	0.7	0.06	50.5	50.9	79.2	94.0	95.5	77.0	77.4	79.6	0.12
FMNIST	2.55	9.9	0.39	96.7	83.5	99.9	100.0	100.0	99.9	99.8	100.0	0.26

Table 1: Two notions in the frozen features (final evaluation images; noise seeds 0–2). $\|\Delta_{180}\|/r_R$: distance between a bank image and its half turn in the space of the kNN term, divided by its distance to its R-th nearest bank image (median). Rot. var.: share (%) of the variance of the head's input features (four rotated views of the bank) explained by the rotation. Var. frame: variance carried by the three linear-discriminant directions of the four rotations, as a multiple of that of three random directions (the top three principal directions carry 329 to 11,688 times more). PC-3 / frame-3: AUROC (%) of a kNN rule restricted to the top three principal / the three discriminant directions (Euclidean, 10th neighbour, 10,000 bank images). Head on $\mathbf{g}$: the head trained on the whole-image statistics only. kNN term, orientation term, sum: the terms of Eq. (2) and Upright kNN. ρ: Spearman correlation of the two terms on natural OOD images (mean over the OOD sets).

ages, so a kNN rule restricted to them detects upside-down images no better than the kNN term does (52.2–77.4%), and a kNN rule in the full standardised head space, which sees the 2×2 grid, is at chance with the Euclidean distance (47.4–50.4%), reaches 56.4–81.0% with the cosine distance, and is at chance again when the space is whitened (Mahalanobis, 45.3–54.7%; supplement). What separates the rotations linearly are directions of tiny within-rotation scatter, which carry little variance (the three discriminant directions, 0.06–1.7 times three random directions): a kNN rule restricted to them (frame-3) detects flips on CIFAR-10, CIFAR-100 and ImageNet-64 (75.2–94.6%), where the top three principal directions (PC-3) do not (50.9–61.3%). A discriminative readout finds such directions on its own: on these four datasets the head reaches 78.6–99.2%, and a head trained on the whole-image statistics $\mathbf{g}$ alone, the very input of the kNN term, still reaches 81.7–99.0%, against 50.5–77.8% for the kNN term on the same $\mathbf{g}$; the grid statistics $\mathbf{q}$ change this by -3.1 to $+1.5$ points (supplement). On the two aligned datasets, CelebA and FashionMNIST, a flipped face or shoe is far from every bank image, and even the Euclidean kNN rule in the full space sees it (92.8–95.9%). The frame is only partly in trivial image statistics: a head on raw-pixel means and standard deviations (whole image and grids) reaches 43.9–64.0% accuracy on the natural-image datasets against 73.1–92.9% from $\mathbf{g}$, though it suffices on the aligned ones (91.3–96.4%; supplement).

Both notions detect distribution shift, on different images. The orientation term is not a flip detector only: on natural OOD sets it reaches 89.9% mean AUROC on its own (kNN term 92.5%), but there the two terms are only weakly rank-correlated ($\rho = 0.10$–0.36 per ID dataset; up to 0.5 on the near-OOD pairs) and flag different images. At the thresholds that keep 95% of the ID images (set on the same images), on ImageNet-64 the kNN term flags 15.6% of the natural OOD images and the orientation term 21.5%, but only 4.3% are flagged by both, close to the 3.3% expected for independent decisions. Their sum is at least as good as the better term on every ID dataset (Table 1), though not on every pair: by the bootstrap rule it falls below the better single term on 5 of the 47 pairs, by up to 5.7 points (the three largest: CIFAR-10→CIFAR-100 80.3 vs 86.0%; ImageNet-64→iNaturalist 71.9 vs 75.1%; CIFAR-100→CIFAR-10 64.1 vs 66.3%). Figure S2 in the supplement shows the joint distribution of the two terms.

What each notion responds to. The images that only one term flags show the difference (Figure S1 in the supplement: per ImageNet-64 OOD set, the image flagged by one term alone that the other finds most ID-like). In those images, the kNN term alone flags colour and texture statistics unusual for ImageNet (underwater scenes, dense vegetation, a dyed fabric, fog) while the head still finds an upright frame; the orientation term alone flags images that the kNN term ranks as typical but whose frame the head does not recognise as upright: close-ups, an object without a canonical pose, trees photographed from below. On CIFAR-10, CIFAR-100 and ImageNet-64 the head's posterior for such images peaks on a wrong angle in 65.2–89.1% of cases and has no angle above 0.5 otherwise: OOD content is read as rotated, not as frameless (supplement). Frame conformity needs familiar content: with CIFAR-10 as ID, it separates CIFAR-100 images better than the kNN term does (86.0% vs 77.4%). It can also be fooled: with CIFAR-100 as ID, CIFAR-10 images look *more* canonical than CIFAR-100 test images (AUROC 28.4%, i.e. the OOD images score higher), which causes the only loss of Upright kNN to its own kNN term (64.1 vs 66.3%).

When the ID data have a canonical frame, and when not. We trained the same kind of U-Net on three further ID datasets (supplement). On chest X-rays (ChestMNIST [Yang *et al.*, 2023a]), which have a strict acquisition frame, the head reaches 99.9% held-out accuracy and the orientation term detects rotated and upside-down radiographs with 100.0% AUROC on each transformation; as on the aligned datasets above, the frame is so strict that the kNN term also sees it (at least 98.6%; likelihood 88.7%, CFS 96.4%). On histology tiles (PathMNIST) and EuroSAT [Helber *et al.*, 2019] aerial tiles, which have no canonical up, the head still learns acquisition cues (75.8% and 72–74% held-out accuracy), and this is where the frame notion misleads: on PathMNIST the orientation term alone is inverted on other modalities (25.5%) and adding it costs the kNN term nearly six points there (92.2% vs 97.9%) and two on natural images (4/0/13 W/T/L over 17 sets); on EuroSAT it changes nothing (99.8% vs 99.8%) but flags rotated tiles (75.8%), unwanted when rotations count as

Method	ID dataset (mean AUROC over its natural OOD sets) CIFAR-10	SVHN	CelebA	CIFAR-100	ImageNet-64	FMNIST	Mean	Near	Flip (probe)	W/T/L of ours	NFE	CPU s/img
Likelihood, score, reconstruction and dimension statistics of the diffusion model												
NLL	58.2	98.9	79.9	53.8	52.3	33.3	62.7	50.7	56.6	43/2/2 (39/1)	256	46
IC (PNG)	73.4	92.1	87.3	66.5	47.1	86.2	75.4	50.0	57.1	47/0/0 (46/0)	256	46
IC (JPEG2K)	71.4	74.1	82.0	65.2	46.4	40.8	63.3	49.3	56.6	47/0/0 (46/0)	256	46
TT	56.8	98.6	85.1	51.4	58.1	61.6	68.6	52.2	53.9	44/2/1 (43/0)	256	46
MSMA*	63.6	99.1	99.7	61.3	63.2	96.4	80.5	53.2	63.3	42/5/0 (32/0)	20	3.5
DiffPath	71.0	99.5	99.9	64.9	54.9	80.2	78.4	51.5	62.3	35/12/0 (28/0)	10	1.5
GEPC	51.9	91.3	99.1	55.6	51.2	84.4	72.3	52.4	52.1	45/2/0 (41/0)	14	2.3
EigenScore	73.4	85.8	86.9	47.6	44.2	94.3	72.0	49.1	75.4	47/0/0 (47/0)	180	27
DDPM-OOD	77.1	**99.9**	99.9	59.2	56.1	94.2	81.1	51.7	69.0	34/9/4 (28/1)	2,475	–
LMD	75.2	83.5	98.4	58.9	40.1	98.1	75.7	48.2	73.4	46/1/0 (42/0)	10,000	–
LID+NLL§	71.6	99.2	87.2	66.2	–	80.6	81.0	–	58.7	37/2/1 (32/0)§	12,544	–
Features of the diffusion model												
CFS (diag.)	73.8	90.7	99.7	65.3	67.2	97.6	82.4	60.8	59.3	43/4/0 (34/0)	1	0.16
CFS (kNN)	84.9	92.1	99.9	72.3	63.9	99.9	85.5	62.8	70.6	40/7/0 (31/0)	1	0.16
kNN (basic)	88.6	98.6	**100.0**	80.3	59.1	99.7	87.7	60.5	68.2	39/8/0 (23/0)	1	0.16
kNN (tuned)	93.7	99.4	**100.0**	85.4	77.0	99.9	92.6	70.0	72.7	24/22/1 (10/1)	1	0.16
Networks trained on the ID images												
Rotation pred.	85.3	95.4	**100.0**	61.8	62.1	99.4	84.0	60.4	94.6	35/12/0 (33/0)	4‡	0.02
CSI§	94.7	97.8	**100.0**	68.5	–	**100.0**	92.2	–	94.5	22/16/2 (21/1)§	40‡	–
Upright kNN (ours)	**95.4**	99.7	**100.0**	**85.5**	**79.6**	**100.0**	**93.4**	**70.9**	86.6	–	1	0.17
kNN term only	93.7	99.4	100.0	85.0	77.0	99.9	92.5	69.9	72.8	28/18/1 (11/1)	1	–
orientation term only	93.8	95.7	100.0	72.9	77.4	99.8	89.9	61.3	94.7	29/14/4 (27/3)	1	–

Table 2: Final evaluation (AUROC, %; 500 images per set). Columns 2–7: mean over the natural OOD sets of each ID dataset; Mean: their average; Near: mean over the four near-OOD pairs (–: not run on ImageNet-64); Flip: mean over the upside-down ID sets (a probe, not an OOD target; unmarked). W/T/L: wins/ties/losses of Upright kNN against the row over the 47 natural pairs by the bootstrap rule, and in parentheses its wins/losses of at least one point. NFE: network evaluations per image, forward and backward passes counted (‡ResNet-18 passes). CPU: seconds per image, CIFAR-10 network, 2 threads. §Not run on all six ID datasets (CSI: the five 32×32 datasets, 40 pairs; LID+NLL: the five 32×32 datasets, 40 pairs (no public configuration for ImageNet-64)); W/T/L over the pairs run. Best and second best per column (single-term rows excluded) are marked **blue** and underlined.

ID. Frame conformity is a property of the data, not of the generator: for five CIFAR-10 generators we trained, without augmentation, with horizontal flips, or with random 90° rotations (probability 0.25 to 1), the head trained on the upright bank keeps 88.9–89.2% accuracy and 97.4–97.9% flip AUROC, whereas the likelihood of every one of them is at chance on flips (50.0–51.0%) and near chance on natural OOD sets (52.2–52.6%), and adding $\log P(\mathrm{up} \mid \mathbf{x})$ to it changes nothing (at most 51.1%; Section 3.3).

6 OOD Detection Results

6.1 Six ID Datasets

Table 2 gives the final evaluation. Upright kNN averages 93.4% AUROC on natural OOD sets, 7.9 points above the best prior detector (CFS with a kNN head, 85.5%); the best detector built on likelihoods, scores, reconstructions or dimension reaches 81.1% (DDPM-OOD); LID+NLL (the five 32×32 datasets) averages 81.0% against our 96.1% there. Against each prior detector run on all six datasets it wins 34–47 of the 47 natural pairs, 28 or more by at least one point, and loses at most 4 (SVHN→CIFAR-10, SVHN→CIFAR-100, SVHN→CelebA, ImageNet-64→iNaturalist). The kNN rule on our features is itself a strong baseline (87.7% basic, 92.6% tuned), so of the 7.9 points over the best prior detector only 0.8 come from the frame term; the other 7.1 come from the kNN rule itself: 2.3 from its head (cosine of unit-normalised statistics instead of CFS's standardised Euclidean distance, on CFS's own descriptors), 1.2 from pooling three decoder blocks, 3.6 from the per-dataset neighbour rank; the block itself adds nothing (supplement, Section S2). Upright kNN improves on its kNN term by 0.9 points on average (28/18/1; 11 wins and 1 loss of at least one point), most on CIFAR-10 (95.4 vs 93.7%) and ImageNet-64 (79.6 vs 77.0%), the two natural-image datasets where the orientation term alone matches or beats the kNN term (Table 1); on the near-OOD pairs it reaches 70.9% against 69.9% for the kNN term and 62.8% for the best prior detector. Its mean FPR95 is 26.5% (tuned kNN 29.8%, best prior detector 43.5%). It takes 0.17 s per image on a CPU, 1/281 of the likelihood and 1/9 of DiffPath.

6.2 CSI, Fusions and the Frame Term as a Plug-in

CSI combines the same two kinds of evidence in a representation trained on ID pixels for 1,000 epochs (36–44 GPU hours per dataset). On the five 32×32 ID datasets it reaches 92.2% mean AUROC on natural OOD sets, against 96.1% for Upright kNN, which trains no representation (W/T/L 22/16/2 over 40 pairs). The comparison is not one-sided: CSI is better on the hardest near-OOD pair, CIFAR-10→CIFAR-100 (91.1% vs 80.3%), and on the probe (94.5% vs 87.9%), but in our run collapses with CIFAR-100 as ID (68.5%; 9.8% on CelebA; supplement), which is the whole gap: on the other four datasets it is within 1.9 points. A rotation classifier trained on pixels does not combine well with the kNN term: fused by z-scores it gives 91.2%, *below* the kNN term alone (92.5%), because its confidence is less aligned with distribution shift (84.0% on natural OOD sets) than our orientation term (89.9%). Fusing our two terms by z-scores or Fisher's method (a scale fitted on extra ID images) gives 93.1% and 93.1%, on the plateau of Section 3.3; dropping d gives 92.0% (supplement). Conversely, the frame term improves

Method	OOD Near	OOD Far	csID (higher = flagged) V2	csID C	csID R	Full spectrum Near	Full spectrum Far
Upright kNN	**70.2**	**79.0**	52.4	92.3	77.9	53.9	63.3
kNN term	69.5	76.4	52.2	92.1	73.4	55.0	61.9
orient. term	63.5	77.4	50.9	80.7	77.8	50.0	64.8
kNN (basic)	60.5	68.1	50.6	83.6	63.9	51.9	60.4
CFS (diag.)	60.7	75.0	52.0	77.3	68.7	51.4	65.1
CFS (kNN)	56.3	71.0	51.5	80.7	67.0	46.8	63.0
Rotation pred.	53.1	70.3	49.7	63.2	65.1	47.1	64.4
NLL	50.4	64.9	51.0	43.3	31.3	**56.5**	**69.3**
TT	56.7	63.4	49.6	66.1	62.7	50.4	56.2
MSMA*	56.7	59.8	46.9	79.1	72.1	46.1	49.8
DiffPath	53.3	62.6	52.1	57.6	69.6	45.7	54.0

Table 3: OpenOOD v1.5 ImageNet-1K at 64×64 with EDM2-XS (AUROC, %; 1,000 images per set; frozen ImageNet-64 detectors). Near: SSB-hard, NINCO; far: iNaturalist, Textures, OpenImage-O; csID: ImageNet-V2, -C, -R against ImageNet val (ID); full spectrum: ID and csID images together as the positive class. Marks as in Table 2; csID columns unmarked; the weaker detectors are in the supplement.

other neighbourhood detectors with the same parameter-free weight, d estimated in their own space (post hoc, expectation written in advance): CFS (kNN) rises from 85.5% to 88.6% (W/T/L 41/5/1) and the basic kNN from 87.7% to 89.8% (34/12/1), more than the 0.9 points it adds to our stronger kNN term, whereas a Gaussian log-density of CFS's descriptors is unchanged (+0.2), as the scale argument predicts. A z-sum of the frame term with each of 12 comparison methods helps every one of them, but exceeds the orientation term alone (89.9%) only for the two neighbourhood detectors (supplement).

6.3 ImageNet-1K (OpenOOD v1.5)

We apply the frozen ImageNet-64 detectors, without refitting, to the full OpenOOD v1.5 ImageNet-1K benchmark [Zhang *et al.*, 2023] at 64×64, whose OOD datasets are those of our ImageNet-64 tier with their standard sets: ID is ImageNet validation [Russakovsky *et al.*, 2015]; near-OOD is SSB-hard [Vaze *et al.*, 2022] and NINCO [Bitterwolf *et al.*, 2023]; far-OOD is iNaturalist [Huang and Li, 2021], Textures [Cimpoi *et al.*, 2014] and OpenImage-O [Wang *et al.*, 2022]; the covariate-shifted ID sets (csID) are ImageNet-V2 [Recht *et al.*, 2019], -C [Hendrycks and Dietterich, 2019] and -R [Hendrycks *et al.*, 2021]. Development and selection images are excluded; 1,000 images per set, every method on the same (Table 3). Upright kNN reaches 70.2% near-OOD and 79.0% far-OOD AUROC against 60.7% and 75.0% for the best prior detector (CFS, diagonal head), wins at least 4 of the five OOD sets against every prior detector run there, and loses only on iNaturalist, to NLL and TT. Both notions contribute (kNN term alone 69.5/76.4%, orientation term alone 63.5/77.4%); a bank 2.5 or 5 times larger changes neither (supplement). ImageNet-V2 is treated as ID (52.4%), whereas corrupted and rendered images are flagged (ImageNet-C 92.3%, ImageNet-R 77.9%), which the full-spectrum protocol [Yang *et al.*, 2023b] counts as errors; there NLL, which ranks such images as *more* likely than clean ones, leads.

Higher resolution. We also apply the frozen ImageNet-64 configuration to ADM [Dhariwal and Nichol, 2021] at 256×256 in pixel space and to the latent models EDM2-XS at 512×512 and DiT-XL/2 [Peebles and Xie, 2023] at 256×256 (null class), every image through the same 224-pixel bottleneck, since the OOD sets come at various resolutions (supplement; protocol written first). With ADM, Upright kNN reaches 83.7% near-OOD and 76.4% far-OOD AUROC against 69.4% and 76.4% for the best prior detectors, but here the gain over 64×64 belongs to the kNN term (84.1% alone), and part of it to how the sets are stored. With the same final resampling for every set (all images through 112 pixels; a post hoc control, expectation written first), SSB-hard keeps 87.7%, but NINCO falls from 80.5 to 69.0% and iNaturalist from 67.4 to 51.9% (chance), leaving 78.4% near-OOD (supplement). With EDM2-XS it reaches 64.7/74.5% against 61.2/72.7% for CFS; with DiT-XL/2 it is weaker (58.3/71.8%) and below its own orientation term: one input-independent channel per block dominates the pooled statistics, so the kNN distance barely resolves images, whereas the head standardises its inputs. In all three models the orientation term detects upside-down images (91.3–96.9%) and the kNN term does not (49.9–51.5%).

7 Limitations and Conclusion

Frame conformity encodes whatever orientation statistics the ID data have. It adds little when the content looks alike upside down (SVHN; flip AUROC 64.5%), can find OOD content more canonical than ID content (CIFAR-100 vs CIFAR-10), and when the ID data have no canonical up the head learns acquisition cues that do not transfer, which costs accuracy on histology tiles and flags rotated aerial tiles; the head's held-out accuracy does not announce this failure (75.8% on histology, 77.3% on ImageNet-64 where the term helps), so the practitioner must know whether the data have a frame. Near-OOD detection remains hard (about 70% AUROC on ImageNet-64; CSI, trained on the ID pixels, is better on CIFAR-10→CIFAR-100). A second kNN term on the deepest decoder block, added post hoc as an ablation (supplement, Section S9; a second use of the final images), lifts this pair to 88.3% and the mean to 94.6% at up to two evaluations, but counts the content density twice, outside Proposition 1. The method presupposes a diffusion model of the ID data, whose training costs far more than CSI's, and inherits its quality: a 39M-parameter U-Net we trained on CIFAR-10 in place of EDM drops the kNN term from 93.7% to 79.5–82.0%; our claim concerns the best use of a given model. The pooled statistics of the kNN term collapse when a few channels carry large input-independent activations (DiT-XL/2); the detector flags covariate shift, penalised by full-spectrum OOD detection.

Within these limits, one frozen denoiser pass holds two notions of ID conformity, closeness to training images and the canonical frame, which disagree per image and which Bayes' rule says how to add; their sum beats 14 prior detectors under a frozen protocol and holds on OpenOOD ImageNet-1K at one evaluation per image.

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